END TO END COFFEE SHOP ANALYTICS

Profitability, Demand & Customer Insight

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Period: 2019 - 2022
Tool: Microsoft Excel + Power Query
Dashboards: 2 Interactive Excel Dashboards
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TABLE OF CONTENT

Section	Page
Cover Page	
Executive Summary	
1. Business Problem	1
2. Objectives	2
3. Methodology	3
4. Dashboard	4
5. Dashboard 2	8
6. Recommendations	11
7. Conclusions	14
8. Appendix	

BUSINESS PROBLEM

The coffee shop had accumulated several years of transactional data but lacked a structured analytical framework to translate that data into strategic decisions. Management could not confidently answer foundational operational questions necessary for profitability management, customer strategy, and demand planning.

1.1 Strategic Questions the Business Could Not Answer

1 Which products were most profitable and which were silently destroying margin?

2 What products and sizes do customers actually prefer?

3 Whether the loyalty programme was driving meaningful additional purchase behaviour?

4 How demand was shifting across years and within months?

1.2 Business Context

Parameter	Detail
Business Type	Specialty coffee shop
Operating Markets	United States, United Kingdom, Ireland
Data Period	January 2019 – August 2022
Total Transactions	3,551 orders
Customer Base	913 unique customers
Primary Tool	Microsoft Excel with Power Query

OBJECTIVES

This analysis was designed to deliver clear, data-supported answers to four strategic questions across three business dimensions (products, demand and customer).

Objective	Business Dimension
1. Identify the most and least profitable coffee types, roast combinations, and product sizes	Product Profitability
2. Assess customer preferences across coffee size, type, and roast to inform potential market expansion decisions	Customer Behaviour
3. Evaluate the impact of the loyalty card programme on purchase frequency and volume	Customer Loyalty
4. Understand demand trends over time to support inventory and staffing planning	Demand Planning

TOOLS & METHODOLOGY

3.1 Analytical Tools

Tool / Technique	Purpose
Microsoft Excel	Data cleaning, transformation, and exploratory analysis
Microsoft Excel	Dashboard design, KPI visualisation, and interactive reporting
Power Query	Data shaping, column standardisation, and data type formatting

3.2 Data Cleaning Steps

The following cleaning operations were performed on the raw dataset prior to analysis:

- Removed duplicate order records identified through order ID cross-referencing
- Standardized date formats across all data tables to enable accurate time-series analysis
- Validated product category labels for consistency, corrected mislabelled roast type entries
- Cross-checked revenue figures against quantity × unit price to identify and correct pricing anomalies

3.3 Known Data Limitations

1. No customer demographic data : Limits depth of customer segmentation; age, gender, and income-level analysis not possible.

2. Incomplete 2022 data (Jan–Aug only): 2022 figures are not representative of a full year. All 2022 data points in trend visuals are marked in grey to flag this limitation.

DASHBOARD1: SALES & PROFITABILITY INTELLIGENCE

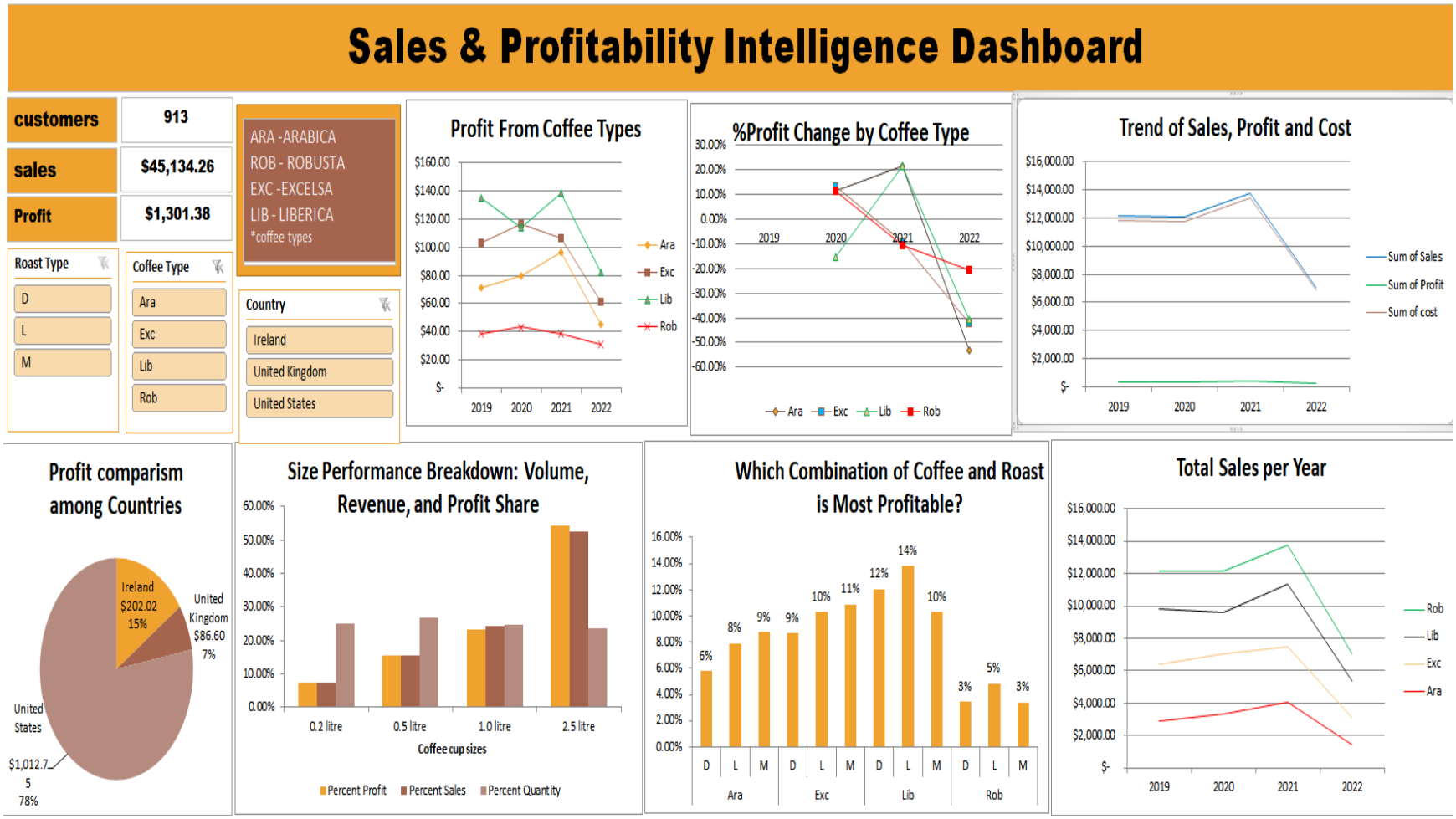
Dashboard Title: *Where Is the Money? A Coffee Profitability & Sales Analysis*

Primary Audience: Senior Management · Finance Team · Product Team

4.1 Visuals Included

Visual	Business Question Answered
Profit Comparison Among Countries	Which markets generate the highest profit and where should growth be focused?
Profit by Coffee Type & Roast Combination	Which product combination should be prioritised for margin optimisation?
% Profit Change by Coffee Type (Year-on-Year)	Which coffee types are improving or deteriorating in profitability?
Size Performance Breakdown (Volume, Revenue, Profit Share)	Does the best-selling size generate the most profit?
Trends of Sales, Profit and Cost Over Time	Is overall revenue growing, declining, or plateauing year on year?
Total Sales Per Year	What is the annual revenue trajectory across the full period?

4.2 Dashboard 1: Sales and Profitability Intelligence Dashboard



4.3 Key Findings

Geographic Performance

The United States accounts for 78% of total profit, indicating heavy market concentration and dependency on a single geography. Ireland contributes 15% of total profit ($202.02) a decent figure given its relative market size. The United Kingdom contributes only 7% of total profit ($86.60), suggesting either low market penetration or poor product-market fit in that market.

Product Profitability

Robusta (ROB) and Excelsa (EXC) recorded a decrease in profitability in 2021 (-10.485% change), while Liberica (LIB) and Arabica (ARA) peaked at 20%+ profitability growth in the same year.Across all three roast types, Robusta has consistently maintained the lowest profit generation. Liberica has consistently delivered the highest profit across all roast types.

- Note: 2022 data points should be treated with caution as the dataset is incomplete, terminating in August 2022.

Coffee Type Sales Performance

Robusta (ROB) is the best-selling coffee type, consistently outperforming all other types by sales volume. Arabica (ARA) is the weakest sales generator, maintaining a record below $4,000 in total sales across the full period. Excelsa and Arabica show relative stability but a declining trend, suggesting potential market fatigue or pricing pressure.

Size Performance Paradox

All four cup sizes show approximately balanced order volumes; demand is evenly distributed across sizes. Despite equal order volumes, the 2.5litre size delivers 30% of total profit, significantly outperforming all other sizes. The 0.2 litre size contributes the least across all three metrics (profit, sales, and quantity) and may not be worth the operational overhead.

Revenue Trend

Revenue grew consistently from 2019 through 2021, with a notable acceleration in 2021. The 2022 revenue appears lower due to the incomplete dataset (data ends August 2022 therefore this is not a genuine decline.) The trend of sales versus profit and cost reveals that profit is not responsive to increases in sales, warranting further investigation into cost structure.

4.4 Profitability Summary Table

Coffee Type	Sales Performance	Profit Performance
Liberica (LIB)	2nd highest sales	HIGHEST profit strongest margins across all roast types
Robusta (ROB)	HIGHEST sales #1	LOWEST profit consistently underperforms on margin
Excelsa (EXC)	Mid-range, declining	Declining profitability, particularly in 2021
Arabica (ARA)	LOWEST sales	Improving 20%+ profit growth in 2021

DASHBOARD 2 — CUSTOMER BEHAVIOUR & DEMAND INTELLIGENCE

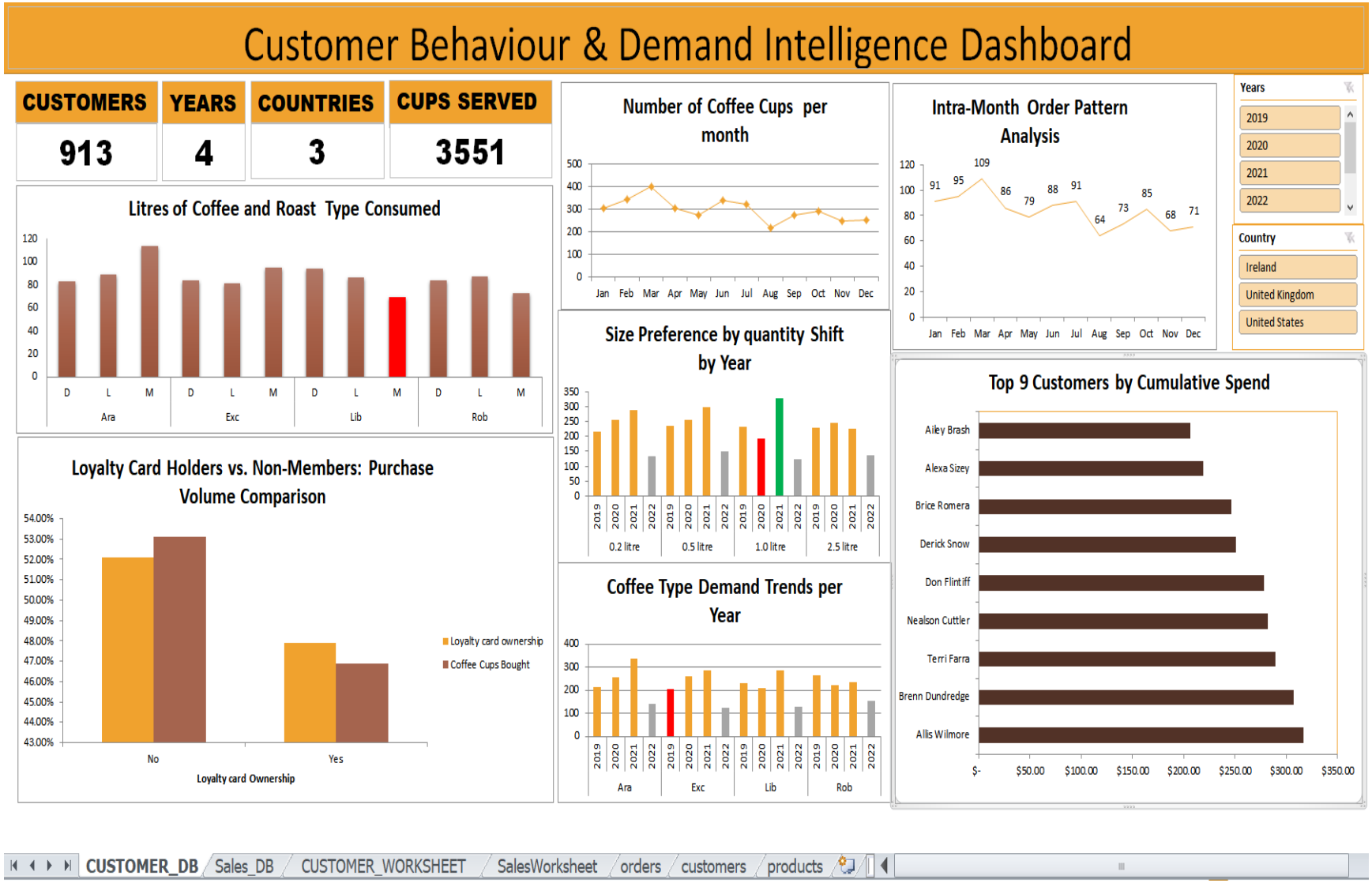
Dashboard Title: *Who Is Buying, What They Want, and When They Come*

Primary Audience: Marketing Team · Operations Team · Customer Success

5.1 Visuals Included

Visual	Business Question Answered
Monthly Order Volume & Intra-Month Pattern	When within a month do orders peak and how can this inform staffing and stock levels?
Size Preference Shifts by Year	Are customers gravitating toward larger or smaller sizes over time?
Coffee Type Demand Trends by Year	Which coffee types are growing in popularity and which are declining?
Loyalty Card Holders vs. Non-Members: Volume	Is the loyalty programme driving meaningfully higher purchase frequency?
Top 9 Customers by Cumulative Spend	Who are the highest-value customers and what is their spend concentration?
Litres of Coffee Consumed by Type & Roast	What is the customer favourite coffee type and roast?

5.2 Dashboard 2:Customer Behaviour and Demand Intelligence Dashboard.



5.3 Key Findings

Demand Seasonality

Monthly order volume peaks between February and March, reaching approximately 400+ cups per month. A significant trough occurs between August and October, where volume drops to approximately 200 cups per month. This predictable seasonal rhythm provides a clear window for targeted promotional activity during the low season. Average orders per month range from 64 to 109, suggesting an average of approximately 3 cups per order.

Size Preference Trends

Demand for 0.2-litre and 0.5-litre cup sizes shows an increasing trend year on year. The 1 litre cup recorded a dip in 2020 but recovered strongly to become the most preferred size by 2021. The 2.5-litre cup has maintained relatively stable demand across the period.

Coffee Type Demand Shifts

Demand for Arabica and Excelsa shows an increasing trend over the period. Liberica and Robusta demand dipped in 2020 but recovered in 2021. These shifting preferences should directly inform procurement decisions and menu prioritisation.

Loyalty Programme Performance

Non-loyalty card holders account for approximately 53% of purchase volume, compared to 47% for loyalty cardholders. This is counterintuitive; the loyalty programme does not appear to be driving meaningfully higher purchase frequency among members. The total number of customers without loyalty cards also exceeds those with cards, suggesting low enrolment rates.

Customer Concentration Analysis

The top customer (Ailey Brash) leads cumulative spend at approximately $320. The top 9 customers are closely clustered between $250 and $340, with no single customer representing a dangerous revenue concentration risk. This is a healthy sign for revenue stability and resilience.

Customer Favourite Drink

The highest volume of coffee consumed by litres was Arabica Light Roast. The lowest volume consumed was Liberica Medium Roast.

RECOMMENDATIONS

The following recommendations are structured to directly address the three critical findings identified in this analysis. Each recommendation is tied to a specific finding and designed for practical implementation.

6.1 Profitability Recommendations

1. Reprice Robusta

Robusta is the highest-volume product yet the lowest-margin product in the range. This is a pricing and cost structure problem, not a demand problem. Conduct a cost-per-unit analysis for Robusta across all roast types and model the impact of a price increase. A modest price adjustment on Robusta, given its inelastic demand as the top seller, could meaningfully shift the overall profit margin without reducing volume.

2. Prioritise Liberica Light Roast

Liberica Light Roast generates the highest profit of any product in the range, accounting for approximately 14% of total profit. Despite this performance, it does not lead in sales volume. It needs visibility, not discounting. Prioritise Liberica Light Roast in menu placement, staff recommendation scripts, and promotional activity. Consider a 'Featured Brew' or 'Best Margin Pick' positioning on the menu to drive organic trial.

3. Investigate the Profit-Sales Decoupling

The trend analysis shows that profit does not respond proportionately to increases in sales, particularly the acceleration in 2021. This decoupling suggests either rising input costs, discounting activity, or a shift in product mix toward lower-margin items. A dedicated cost-structure review covering 2019- 2012 is recommended as a priority action for the finance team.

6.2 Size Strategy Recommendations

4. Commercially Activate the 2.5-Litre Size

The 2.5-litre size delivers 30% of total profit despite receiving no apparent pricing or promotional advantage. This product is not underperforming, it is underutilized. The opportunity is in redirecting existing demand, not generating new demand.

Recommended actions:

1. Display the 2.5-litre size prominently as the 'Best Value' option on the menu and at the point of sale. Introduce an upgrade deal: 'Move from 1 litre to 2.5 litre for just $X more.'

2. Award loyalty members additional points specifically for purchasing the 2.5-litre size. Review whether the 0.2 litre sizes warrants continued stocking given its consistently low contribution across profit, sales, and quantity metrics.

6.3 Loyalty Programme Recommendations

5. Redesign the Loyalty Programme

The current programme is not driving higher purchase frequency among cardholders which is the fundamental KPI for any loyalty scheme. The redesign should focus on two levers: converting non-cardholders and increasing spend among existing cardholders.

Recommended actions:

1. Introduce a sign-up incentive: first drink free, or a bonus points award on registration.
2. Tie loyalty rewards specifically to high-margin behaviours: purchasing 2.5-litre sizes and Liberica products.
3. Set a measurable target: loyalty cardholders should account for at least 55% of purchase volume within months of programme relaunch.

6.4 Demand & Seasonal Recommendations

6 Plan Around the Seasonal Trough

Order volume falls significantly between August and October. This is a predictable and addressable pattern. Launch a targeted promotional campaign in July to capture customers before the dip, and extend it through September to cushion the trough. Use the peak season (February–March) to test pricing changes and new product introductions when footfall is highest.

6.5 Recommendations Priority Matrix

Recommendation	Priority	Responsible Team
Reprice Robusta	High (Immediate)	Finance / Product
Promote Liberica Light Roast	High (Immediate)	Marketing
Investigate profit-sales decoupling	High (Immediate)	Finance
Activate 2.5-litre commercially	Medium (30 days)	Operations / Marketing
Redesign loyalty programme	Medium (60 days)	Marketing / Customer Success
Seasonal promotional calendar	Low (Next quarter)	Marketing

CONCLUSIONS

This analysis set out to answer four questions that the business could not previously answer with confidence: which products were most profitable, what customers preferred, whether the loyalty programme was working, and how demand was shifting over time. All four questions now have clear, data-supported answers.

The most consequential finding is **structural**. Across 3,551 orders and four years of trading, the highest-selling coffee type generated the lowest profit, and the highest-profit coffee type was not the most promoted. Robusta's dominance in sales volume masked a profitability problem that would have remained invisible without this analysis. Liberica, particularly the Light Roast, consistently delivered the strongest margins but never received the commercial attention its performance warranted.

The size analysis reinforces this pattern. All four cup sizes attract roughly equal order volumes, yet the 2.5-litre size contributes 30% of total profit despite receiving no apparent pricing or promotional advantage. This is not a product that is underperforming it is a product that is being underutilized. The opportunity is not to generate new demand but to redirect existing demand toward a higher-margin option that customers are already willing to purchase.

The customer and demand analysis adds important operational context. Order volumes peak between February and March and fall significantly between August and October, providing a predictable seasonal rhythm the business can plan staffing, stock, and promotions around. The top nine customers are closely clustered in cumulative spend between $250 and $340, with no single customer representing a dangerous concentration risk; a healthy sign for revenue stability.

The loyalty programme finding is the clearest immediate call to action. When customers without loyalty cards purchase at a higher volume than cardholders, the programme has failed its primary purpose. Tying loyalty rewards specifically to 2.5-litre purchases and Liberica orders would simultaneously fix the programme's underperformance and steer customer behaviour toward higher-margin products.

The 2.9% Profit Margin - Context and Outlook

The 2.9% overall profit margin is a warning signal, not a crisis. The business demonstrates genuine market traction across 3,551 orders from 913 customers over four years. The problem is not demand , it is a pricing and product mix problem, and both are within management's control to address. Repricing Robusta, promoting Liberica Light Roast, and redesigning the loyalty programme around margin-positive behaviour are three targeted actions that could meaningfully lift profitability without requiring a single new customer.

APPENDIX

8.1 Dataset Information

Attribute	Detail
Dataset Name	coffeeOrdersData
Source	Mochens Github account
URL / Reference	https://github.com/mochen862/excel-project-coffee-sales/blob/main/coffeeOrdersData.xlsx
License	Public
Date Accessed	02-03-2026

8.2 Glossary

Term	Definition
ROB	Robusta - a coffee type known for strong, bold flavour and high caffeine content
LIB	Liberica - a rarer coffee variety; in this dataset the highest profit-margin type
ARA	Arabica - the most common specialty coffee type globally; lowest sales in this dataset
EXC	Excelsa - a variety of Liberica; mid-range in both sales and profitability
Profit Margin	Profit ÷ Sales × 100. The overall margin in this dataset is ~2.9%

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This report was produced as part of a personal data analytics portfolio. All findings and recommendations are based solely on the dataset provided and are intended to demonstrate analytical thinking, data storytelling, and dashboard design capability.